

Identification and Characterization of KMT2C Variants in Breast Cancer Patients

INTERNSHIP REPORT

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The results embodied in this thesis have not been submitted to any other university or institute for the award of any degree or diploma.

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The results embodied in this thesis have not been submitted to any other university or institute for the award of any degree or diploma.

Dr. K. Venkateswar Reddy

DECLARATION

We hereby declare that the project work embodied in the dissertation entitled **"IDENTIFICATION AND CHARACTERIZATION OF KMT2C VARIANTS IN BREAST CANCER PATIENTS"** is being submitted to CBT, UCESTH, JNTUH was carried out by us at Centre for Biotechnology (CBT), University College of Engineering Science and Technology Hyderabad (UCESTH), under the supervision of **Dr. K. Venkateswar Reddy, Assistant Professor, Centre for Biotechnology, University College of Engineering Science and Technology Hyderabad, JNTUH**. This report is submitted in partial fulfillment for the award of Bachelor of Technology in Biotechnology at CBT, UCESTH, Jawaharlal Nehru Technological University Hyderabad, Kukatpally – 500 085, Hyderabad, Telangana State, India.

This work is original and no part of this has been submitted to any other university or institute for the award of any degree or diploma.

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ABSTRACT

Breast cancer is the most commonly diagnosed malignancy among women worldwide and represents a significant public health challenge due to its high incidence and mortality rates. Genetic and epigenetic alterations play critical roles in breast cancer initiation, progression, and therapeutic response. Among epigenetic regulators, the Lysine Methyltransferase 2C (KMT2C) gene has emerged as one of the most frequently mutated chromatin-modifying genes in breast cancer. KMT2C functions as a tumor suppressor by regulating enhancer activity, gene expression, DNA damage repair, and genomic stability. Despite its recognized importance, the mutational spectrum of KMT2C remains relatively underexplored in the Indian breast cancer population.

The present study aimed to identify and characterize sequence variants in the KMT2C gene among breast cancer patients. A total of 9 histopathologically confirmed breast cancer tissue samples and 8 healthy control blood samples were included in the study. Genomic DNA was isolated using the Phenol–Chloroform–Isoamyl Alcohol extraction method. A selected region of the KMT2C gene was amplified by Polymerase Chain Reaction (PCR) using gene-specific primers, generating a 378 bp amplicon. Amplified products were verified by agarose gel electrophoresis and subsequently subjected to Sanger sequencing. Sequence chromatograms were analyzed using CodonCode software, and detected variants were annotated using NCBI dbSNP and ClinVar databases to assess their clinical significance.

Successful PCR amplification and high-quality Sanger sequencing data enabled the identification and characterization of nucleotide variations within the targeted KMT2C region. The study provides valuable insights into the molecular profile of KMT2C in breast cancer patients and highlights the significance of epigenetic regulator genes in breast cancer pathogenesis. These findings contribute to the molecular characterization of breast cancer and support the potential utility of KMT2C variants as biomarkers for disease susceptibility, prognosis, and future precision-medicine approaches.

Keywords: Breast Cancer, KMT2C, Lysine Methyltransferase 2C, Epigenetics, Histone Methylation, Tumor Suppressor Gene, Genetic Variants, Mutation Analysis, DNA Damage Repair, Genomic Stability, Polymerase Chain Reaction (PCR), Sanger Sequencing, Biomarkers, Precision Medicine, Molecular Genetics.

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INTRODUCTION

INTRODUCTION

Breast cancer remains the most frequently diagnosed cancer among women globally, ranking as the second most common cancer overall behind lung cancer and the fourth leading cause of cancer death worldwide. According to the latest GLOBOCAN estimates from the International Agency for Research on Cancer (IARC), approximately 2.4 million new cases are diagnosed annually, accounting for nearly 12% of all cancer diagnoses. The disease claims roughly 694,000 lives each year, making up about 7% of all cancer-related deaths. While it predominantly affects women—ranking as the most common female cancer in 164 out of 186 countries—it also impacts men, who account for roughly 0.5% to 1% of all cases^[1,2].

The global data reveals stark survival inequalities driven by socioeconomic factors and varying healthcare infrastructures. In countries with a very high Human Development Index (HDI), about 1 in 12 women will be diagnosed with breast cancer in their lifetime, but only 1 in 71 will die from it due to robust early screening and advanced treatment options. Conversely, in low-HDI countries, although a lower diagnostic rate of 1 in 27 women is reported due to limited screening access, a disproportionately high 1 in 48 women die from the disease. Driven by population aging and growth, long-term public health projections warn that global breast cancer cases are expected to scale rapidly by 2050, with transitioning economies in Asia and Africa bearing the sharpest increases in mortality^[1].

In Telangana, breast cancer has emerged as the most common malignancy among women and represents a growing public health concern. Data from the Hyderabad Population-Based Cancer Registry (PBCR) under the National Cancer Registry Programme (NCRP) indicate that Hyderabad has one of the highest breast cancer incidence rates in India, with an age-adjusted incidence rate of 48 per 100,000 women. The increasing disease burden has been associated with factors such as urbanization, lifestyle changes, delayed childbirth, and improved cancer detection in India^[3,4].

Breast cancer is a highly heterogeneous disease with considerable variation in its histopathological characteristics, molecular alterations, treatment response, and clinical outcome. Based on the expression of estrogen receptor (ER), progesterone receptor (PR), and human epidermal growth factor receptor 2 (HER2), breast cancer is classified into four major molecular subtypes: Luminal A, Luminal B, HER2-positive, and Triple-Negative Breast

Cancer (TNBC). These subtypes differ substantially in their biological behavior, prognosis, and therapeutic management. Hormone receptor-positive tumors generally respond well to endocrine therapy and have favorable clinical outcomes, whereas HER2-positive tumors benefit from HER2-targeted therapies. In contrast, TNBC lacks ER, PR, and HER2 expression, exhibits aggressive clinical behavior, has a greater propensity for metastasis, and remains associated with poor prognosis due to limited targeted treatment options. The remarkable molecular diversity of breast cancer highlights the importance of identifying reliable biomarkers that can predict disease progression and guide personalized therapeutic strategies^[3–5].

Breast cancer is classified into four major molecular subtypes (**Fig 1.1**).

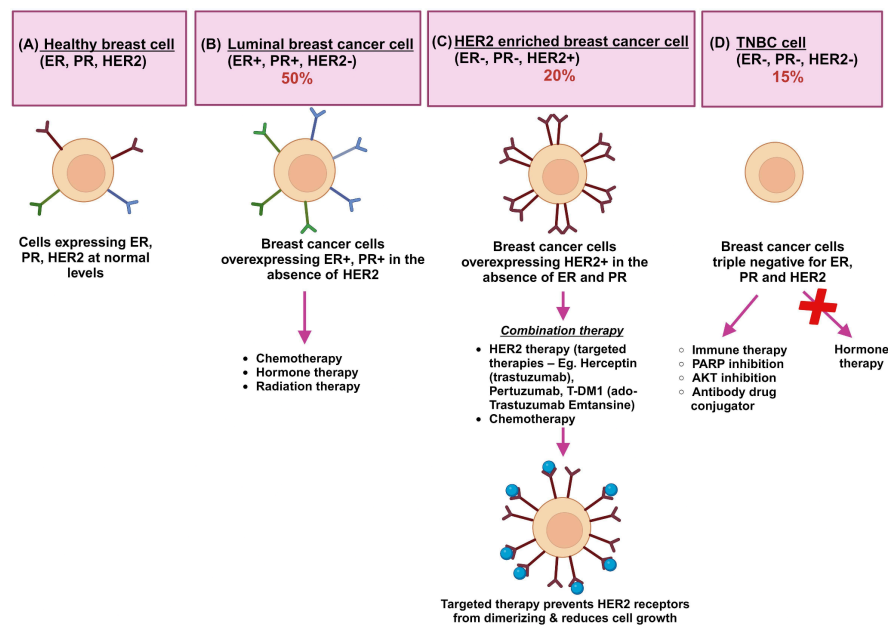


Fig 1.1 Molecular classification of breast cancer based on the expression of estrogen receptor (ER), progesterone receptor (PR), and human epidermal growth factor receptor 2 (HER2), highlighting the major molecular subtypes and their clinical characteristics^[5–8].

The development of breast cancer is driven by the accumulation of driver mutations in oncogenes and tumor suppressor genes that regulate cell proliferation, apoptosis, DNA repair, and genomic stability. Well-characterized genetic alterations include mutations in *TP53*, *PIK3CA*, *BRCA1*, *BRCA2*, *GATA3*, and *ERBB2*, which contribute to tumor initiation and progression. However, genetic mutations alone do not fully explain the complexity of breast cancer. Increasing evidence suggests that epigenetic alterations play an equally important role

in regulating gene expression without altering the DNA sequence. Epigenetic regulation involves DNA methylation, histone modifications, and chromatin remodeling, all of which influence chromatin accessibility and transcriptional activity. Dysregulation of these processes can result in inappropriate activation of oncogenes or silencing of tumor suppressor genes, thereby promoting carcinogenesis. Consequently, epigenetic dysregulation is now recognized as one of the hallmarks of breast cancer^[9–12].

Among the various epigenetic mechanisms, histone modifications are critical regulators of transcriptional activity. Histone proteins undergo several post-translational modifications, including methylation, acetylation, phosphorylation, and ubiquitination, which collectively determine chromatin structure and gene expression. Histone H3 lysine 4 monomethylation (H3K4me1) is a characteristic mark of active enhancers and plays a crucial role in controlling transcription of genes involved in cellular differentiation and development. These modifications are mediated by histone-modifying enzymes, particularly histone methyltransferases, which regulate enhancer activity and maintain normal gene expression patterns. Alterations in these enzymes can disrupt chromatin organization and transcriptional regulation, contributing to tumor development, progression, therapeutic resistance, and metastasis. Consequently, histone methyltransferases have emerged as important molecular regulators and promising therapeutic targets in breast cancer research^[13–16].

One of the most frequently altered histone methyltransferases in breast cancer is *KMT2C* (also known as *MLL3*), a member of the lysine methyltransferase 2 (KMT2) family. *KMT2C* catalyzes histone H3 lysine 4 monomethylation (H3K4me1) at enhancer regions and plays a crucial role in regulating chromatin accessibility and the transcription of genes involved in cell proliferation, differentiation, DNA repair, and hormone receptor signaling. Large-scale genomic studies have identified *KMT2C* as one of the most frequently mutated epigenetic regulators in breast cancer, with variants occurring across different molecular subtypes. Most reported alterations include truncating, frameshift, nonsense, and missense variants that are predicted to impair protein function, supporting the role of *KMT2C* as a tumor suppressor gene. Alterations in *KMT2C* have been shown to disrupt enhancer-mediated gene regulation and contribute to the molecular pathogenesis of breast cancer^[8–12].

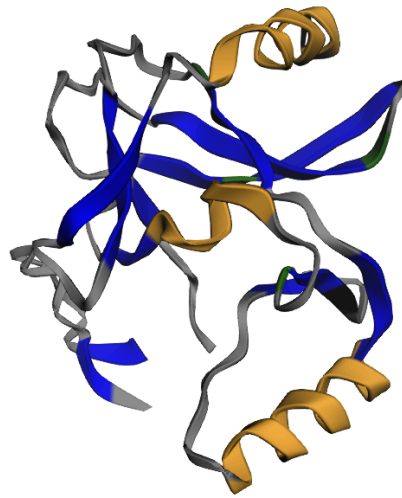


Fig 1.2 : 3D structure of KMT2C protein highlighting its secondary structures

KMT2C exhibits a broad spectrum of genetic alterations in breast cancer, including missense, nonsense, frameshift, splice-site, insertion/deletion (indel), and truncating variants. These variants are distributed throughout the gene, although many cluster within functionally important domains involved in chromatin remodeling and enhancer regulation. Among these, truncating mutations—including nonsense and frameshift variants—are the most frequently reported and are generally associated with loss of function of the *KMT2C* protein. Loss-of-function mutations impair histone H3 lysine 4 monomethylation (H3K4me1) at enhancer regions, leading to dysregulated gene expression, altered hormone receptor signaling, defective DNA repair, and enhanced tumor progression. In contrast, the functional significance of many missense variants remains uncertain and is often classified as variants of uncertain significance (VUS), emphasizing the need for further molecular characterization and clinical interpretation of *KMT2C* variants in breast cancer^[21–24].

Large-scale genomic studies have substantially expanded the understanding of the mutational landscape of *KMT2C* in breast cancer. Comprehensive analyses from The Cancer Genome Atlas (TCGA), the Molecular Taxonomy of Breast Cancer International Consortium (METABRIC), and cBioPortal have identified *KMT2C* as one of the most frequently altered epigenetic regulator genes in breast cancer, with mutation frequencies ranging from approximately 10% to 20%, depending on the molecular subtype and patient cohort. The majority of these alterations are loss-of-function mutations, including frameshift, nonsense, splice-site, and deleterious missense variants, which are distributed across the coding sequence rather than being confined to a single mutational hotspot. Many truncating

mutations disrupt the C-terminal SET methyltransferase domain or other conserved functional domains, resulting in impaired enhancer-mediated histone H3K4 monomethylation and dysregulated gene transcription. The availability of these large genomic datasets has provided valuable insights into the prevalence, distribution, and potential clinical significance of *KMT2C* variants, thereby facilitating their evaluation as biomarkers for breast cancer prognosis and therapeutic response^[5,15,25,26].

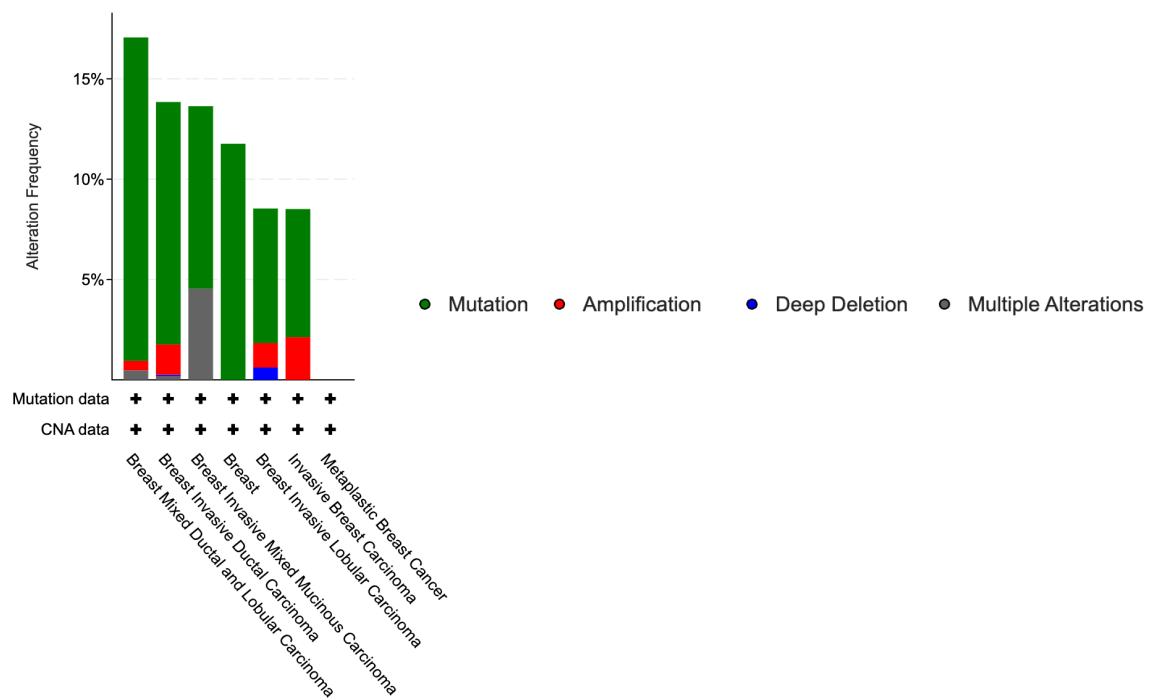


Fig 1.3. Frequency and types of *KMT2C* genomic alterations across different breast cancer subtypes obtained from the TCGA/cBioPortal database. Somatic mutations constitute the predominant genomic alteration, while amplifications, deep deletions, and multiple alterations occur less frequently.^[5,26]

Analysis of TCGA and cBioPortal datasets demonstrates that *KMT2C* is recurrently altered across multiple breast cancer subtypes, with alteration frequencies reaching approximately 17% in mixed ductal and lobular carcinoma and 12–14% in invasive ductal carcinoma. Most genomic alterations are somatic mutations, supporting the role of *KMT2C* as one of the most frequently mutated epigenetic regulator genes in breast cancer^[5,24].

Recent advances in next-generation sequencing (NGS) and Sanger sequencing have enabled the comprehensive detection and characterization of *KMT2C* variants in breast cancer patients. However, despite the increasing number of reported variants, the clinical

significance and pathogenicity of many alterations remain uncertain, with a substantial proportion being classified as variants of uncertain significance (VUS). Accurate interpretation of these variants through established databases such as NCBI ClinVar and dbSNP is essential for distinguishing pathogenic mutations from benign polymorphisms and understanding their potential clinical relevance. Although numerous *KMT2C* variants have been identified in global breast cancer cohorts, studies investigating their frequency and spectrum in Indian breast cancer patients remain limited. Since genetic variation differs across populations, population-specific characterization of *KMT2C* variants is necessary to improve variant interpretation, enhance our understanding of the mutational landscape of breast cancer, and support the development of precision medicine approaches in the Indian population^[15,18,22].

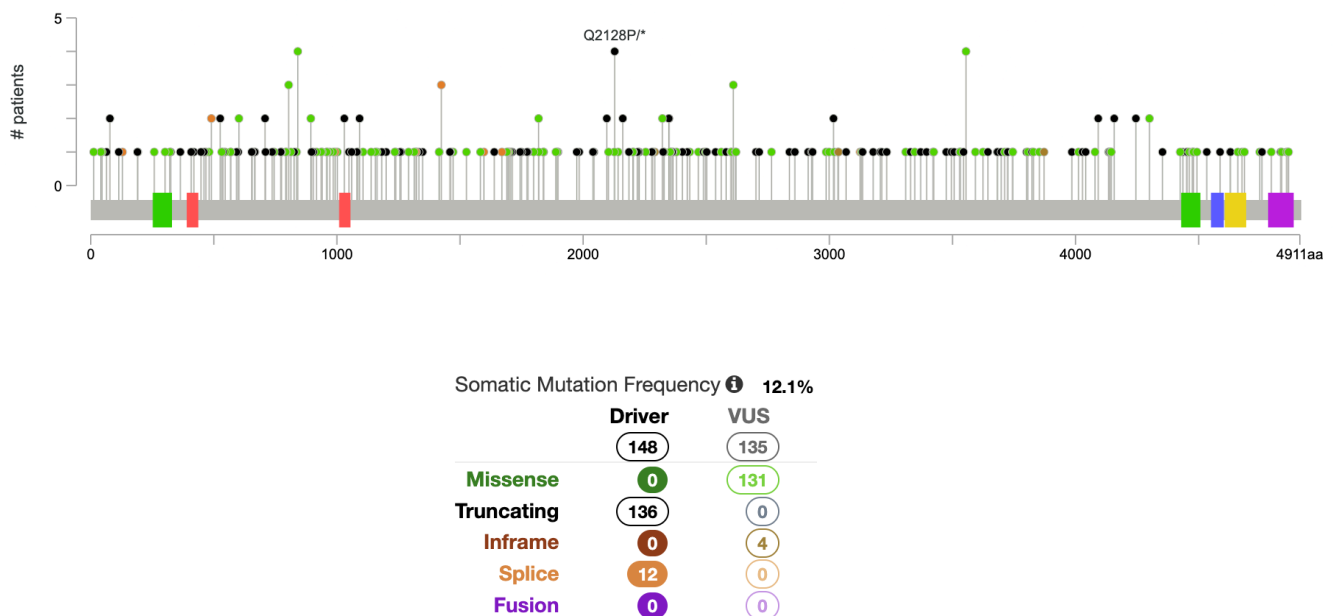


Fig 1.4 Distribution of *KMT2C* mutations across functional protein domains in breast cancer patients from the TCGA/cBioPortal database. Mutations are distributed throughout the coding region, with numerous truncating and missense variants affecting conserved functional domains, including the PHD zinc finger domains and the C-terminal SET methyltransferase domain^[5,15,24].

Unlike genes such as *TP53* and *PIK3CA*, *KMT2C* does not exhibit a single dominant mutational hotspot. Instead, mutations are distributed throughout the coding sequence, with many truncating variants disrupting conserved functional domains, particularly the SET methyltransferase domain, resulting in loss of histone methyltransferase activity and impaired enhancer regulation^[5,15,24].

Despite the increasing incidence of breast cancer in India, genomic studies investigating *KMT2C* variants in Indian breast cancer patients remain limited. Population-specific genetic profiling is essential because allele frequencies, variant spectra, and pathogenicity may differ among ethnic groups, influencing disease susceptibility and clinical outcomes. Therefore, identifying and characterizing *KMT2C* variants in the local population is important for improving the understanding of breast cancer genetics and supporting the development of precision medicine approaches^[22,25].

Therefore, this study aims to identify and characterize *KMT2C* gene variants in breast cancer patients using Sanger sequencing. It seeks to determine the frequency of these variants in the study population and evaluate their pathogenicity and clinical significance through publicly available databases such as NCBI ClinVar and dbSNP. By analyzing the identified variants, this study aims to improve our understanding of the mutational profile in breast cancer and provide insights into its potential role in breast cancer genetics, thereby supporting future research and clinical investigations.

CHAPTER 2

REVIEW OF LITERATURE

REVIEW OF LITERATURE

2.1 Identification of KMT2C as a Recurrently Altered Gene in Breast Cancer

The molecular characterization of breast cancer has undergone substantial refinement following the advent of large-scale sequencing initiatives. Early genomic investigations predominantly focused on canonical driver genes including *BRCA1*, *BRCA2*, *TP53*, and *PIK3CA*, owing to their established roles in genomic maintenance and oncogenic signaling. However, comprehensive sequencing studies subsequently revealed that genes involved in chromatin remodeling and epigenetic regulation also constitute a significant component of the breast cancer mutational landscape^[5,10,12].

One of the landmark studies that advanced this understanding was provided by *Pereira et al. (2016)*, who performed targeted sequencing of 2,433 primary breast tumors and identified 40 significantly mutated driver genes. Among these, *KMT2C* emerged as one of the most frequently altered genes, exhibiting mutations in approximately 11.4% of analyzed tumors. Notably, *KMT2C* ranked among the few genes demonstrating mutation frequencies exceeding 10%, thereby implicating epigenetic deregulation as a fundamental event in breast tumorigenesis rather than a secondary consequence of disease progression^[5,26].

Subsequent investigations utilizing The Cancer Genome Atlas (TCGA) and METABRIC cohorts consistently confirmed the recurrent alteration of *KMT2C* across multiple breast cancer subtypes. These studies collectively demonstrated that *KMT2C* is not merely an infrequently mutated chromatin-associated gene but rather represents one of the most commonly disrupted epigenetic regulators in breast cancer. Furthermore, recurrent identification of *KMT2C* across independent cohorts suggested that alterations affecting chromatin architecture may contribute significantly to disease heterogeneity, therapeutic response, and clinical outcome^[5,15,23].

The repeated observation of *KMT2C* alterations in large patient cohorts shifted scientific interest toward understanding the functional implications of its disruption. Consequently, *KMT2C* has become increasingly recognized as a candidate tumor suppressor gene whose

inactivation may influence transcriptional regulation, enhancer activity, and breast cancer progression^[15,18,21].

2.2 Spectrum of KMT2C Variants Reported in Breast Cancer

Following the identification of *KMT2C* as a recurrently altered gene in breast cancer, subsequent studies focused on characterizing the nature and distribution of individual genetic alterations occurring within the gene. Unlike classical oncogenes that frequently harbor a limited number of activating hotspot mutations, *KMT2C* exhibits a highly heterogeneous mutational profile comprising diverse loss-of-function variants distributed throughout its coding sequence^[15,18,21].

Comprehensive mutation profiling performed by *Lee et al.* demonstrated that *KMT2C* alterations in breast cancer include missense mutations, nonsense mutations, frameshift insertions, frameshift deletions, splice-site variants, and in-frame insertions. Analysis of 958 breast cancer samples identified 80 distinct *KMT2C* mutations, with truncating genomic changes constituting a substantial proportion of the observed events. The predominance of protein-disrupting mutations suggested that loss of *KMT2C* activity, rather than gain-of-function mechanisms, represents the principal biological consequence of *KMT2C* alteration in breast cancer.

Similarly, analyses derived from TCGA and metastatic breast cancer cohorts revealed that frameshift, truncating, and missense variants collectively account for the majority of *KMT2C* mutations. Importantly, many of these mutations occur upstream of the catalytic SET domain, potentially abolishing histone methyltransferase activity and impairing enhancer-mediated transcriptional regulation^[15,21,23].

The heterogeneous distribution of *KMT2C* variants observed across independent datasets indicates that functional disruption of the gene may arise through multiple mutational mechanisms. This observation provides a strong rationale for variant-level investigation, particularly in studies aiming to determine whether specific categories of *KMT2C* alterations are associated with distinct clinical outcomes.

2.3 KMT2C Alterations Across Human Malignancies and Their Relevance to Breast Cancer

The identification of recurrent *KMT2C* alterations in breast cancer prompted investigations into its prevalence across other malignancies. Large-scale pan-cancer analyses subsequently demonstrated that *KMT2C* is among the most frequently mutated epigenetic regulator genes in human cancer, highlighting its broad biological significance^[5,15,28].

Alterations affecting *KMT2C* have been reported in endometrial carcinoma, bladder cancer, colorectal cancer, gastric cancer, lung cancer, and several hematological malignancies. Despite this widespread distribution, mutation frequencies vary considerably among tumor types, reflecting differences in tissue-specific transcriptional programs and selective evolutionary pressures during tumor development^[28,29].

Within breast cancer specifically, *KMT2C* mutation frequencies ranging from approximately 7% to 14.5% have been reported across TCGA, METABRIC, and independent clinical cohorts. *Ning et al.* reported mutation frequencies of 8.0%, 7.0%, and 14.5% in Chinese, TCGA, and METABRIC breast cancer cohorts, respectively, thereby confirming the recurrent nature of *KMT2C* alterations across geographically distinct populations^[5,23,30].

Several studies have demonstrated that *KMT2C* alterations are not restricted to breast cancer but occur recurrently across a wide range of human malignancies. The Cancer Genome Atlas (TCGA) Research Network identified recurrent frameshift, nonsense, and splice-site *KMT2C* mutations in colorectal carcinoma, indicating that loss-of-function variants represent a major mechanism of *KMT2C* inactivation. *Rao and Dou (2015)* comprehensively reviewed the mutational landscape of the *KMT2* family and reported recurrent *KMT2C* alterations in colorectal, bladder, endometrial, lung, and breast cancers, highlighting that protein-truncating variants predominate across multiple tumor types. In pancreatic ductal adenocarcinoma, genomic studies have identified recurrent somatic *KMT2C* mutations and deletions, supporting its proposed role as a tumor suppressor in pancreatic tumorigenesis. Furthermore, *Cho et al. (2018)* demonstrated that diffuse-type gastric adenocarcinoma harbors recurrent *KMT2C* mutations, including missense, nonsense, and splice-site variants, and further showed that loss of *KMT2C* promotes epithelial-to-mesenchymal transition and is associated with poor clinical outcome. Collectively, these findings indicate that recurrent *KMT2C* variants

represent a common molecular event across multiple solid tumors, thereby providing a strong rationale for investigating their biological and clinical significance in breast cancer^[28,29,32,33].

Beyond its mutation frequency, several biological observations further justify the selection of *KMT2C* as a candidate gene in breast cancer. Several functional studies have demonstrated that *KMT2C* alterations are associated with estrogen receptor signaling, endocrine therapy response, and transcriptional dysregulation in breast cancer, thereby highlighting its clinical relevance. Consequently, breast cancer represents a particularly informative model for investigating *KMT2C*-associated molecular alterations and their potential clinical significance.

2.4 Mutational Spectrum of *KMT2C* in Breast Cancer

Accumulating evidence indicates that *KMT2C* alterations in breast cancer arise predominantly through somatic mutational events acquired during tumor evolution. These alterations encompass missense, nonsense, frameshift, splice-site, and deletion mutations, many of which are predicted to generate truncated proteins lacking essential functional domains. The predominance of loss-of-function variants strongly supports the classification of *KMT2C* as a tumor suppressor gene^[15,21,31].

In contrast, germline alterations involving *KMT2C* appear to be considerably less common. Although germline pathogenic variants have been documented in cancer predisposition studies, their contribution to hereditary breast cancer susceptibility remains less clearly defined compared with established susceptibility genes such as *BRCA1* and *BRCA2*. Nevertheless, the presence of inherited *KMT2C* variants suggests that constitutional disruption of epigenetic regulatory pathways may contribute to individual variation in cancer risk^[29,34].

A notable characteristic of *KMT2C* mutational architecture is the absence of highly recurrent hotspot mutations. Unlike classical oncogenes such as *PIK3CA* and *KRAS*, or the tumor suppressor *TP53*, which harbor recurrent hotspot mutations concentrated at specific codons, *KMT2C* alterations are distributed throughout the coding sequence without a consistently recurrent hotspot. Many reported variants cluster within regions preceding the SET domain,

resulting in premature protein truncation and loss of catalytic activity; however, no single dominant hotspot has been consistently identified across breast cancer cohorts^[15,29,31].

The dispersed distribution of *KMT2C* mutations indicates that tumor development is associated with overall loss of gene function rather than recurrent alterations affecting specific amino acid residues. Therefore, the absence of recurrent hotspot mutations highlights the importance of comprehensive sequencing approaches capable of identifying the full spectrum of *KMT2C* variants present in breast cancer^[15,29,31].

The biological significance of individual *KMT2C* variants extends beyond their mere presence within the gene. Depending on their location and predicted molecular consequence, variants may alter protein structure, disrupt interactions with chromatin-associated regulatory complexes, or impair enhancer-mediated transcriptional regulation. Such alterations can modify transcriptional regulation of downstream target genes, ultimately influencing cellular processes including proliferation, differentiation, and therapeutic response. These observations have prompted increasing interest in functional characterization of individual *KMT2C* variants^[15,18,29].

The characterization of individual *KMT2C* variants has become an important area of investigation because accumulating evidence indicates that individual variants may differ substantially in their molecular consequences, functional impact and clinical relevance. While numerous somatic variants have been identified through large-scale sequencing studies, the pathogenic significance of many variants remains uncertain, and their associations with tumor behavior, therapeutic response, and patient prognosis have not been fully established. Accordingly, variant-level analyses have become an integral component of breast cancer genomics research, enabling discrimination between clinically relevant alterations and variants of uncertain significance. These observations highlight the need for continued investigation of *KMT2C* variants in diverse patient populations, thereby providing a strong rationale for studies aimed at identifying and evaluating their potential clinical significance in breast cancer.

2.5 Role Of KMT2C as a Tumor Suppressor in Breast Cancer

Following the characterization of *KMT2C* variants and their distribution across breast cancer cohorts, it is essential to understand the biological significance of these alterations. Previous sequencing studies consistently demonstrated that the majority of *KMT2C* alterations comprise protein-disrupting variants, prompting investigations into whether *KMT2C* functions as a tumor suppressor in breast cancer. Unlike oncogenes, which promote tumorigenesis through activating mutations, tumor suppressor genes contribute to cancer development primarily through loss of normal cellular functions. Consequently, the mutational profile of *KMT2C* provides important clues regarding its role in breast cancer pathogenesis^[15,18,29,31].

Large-scale genomic studies have consistently demonstrated that the majority of *KMT2C* alterations are predicted to impair protein function. *Pereira et al. (2016)* identified *KMT2C* among the significantly mutated genes in breast cancer and reported a predominance of protein-disrupting variants. Similarly, analyses from The Cancer Genome Atlas (TCGA) and METABRIC cohorts revealed recurrent loss-of-function alterations affecting multiple functional regions of the gene. The widespread occurrence of truncating mutations suggested that selective pressure during tumor evolution favors inactivation of *KMT2C* rather than activation of specific residues, a characteristic commonly observed in tumor suppressor genes^[5,23,26,31].

Experimental evidence further supports the tumor suppressive role of *KMT2C*. Functional studies have demonstrated that depletion of *KMT2C* results in altered enhancer activity, disruption of transcriptional regulation, and enhanced cellular proliferation. *Gala et al.* reported that loss of *KMT2C* promotes oncogenic transcriptional programs and contributes to endocrine therapy resistance in estrogen receptor-positive breast cancer. These findings indicate that *KMT2C* is required for maintaining transcriptional networks that restrict uncontrolled cellular growth and preserve normal cellular identity^[15,18,35].

In addition to transcriptional dysregulation, emerging evidence suggests that *KMT2C* contributes to the maintenance of genomic stability. Deficiency of *KMT2C* has been associated with impaired DNA damage repair mechanisms and increased genomic instability,

thereby facilitating the accumulation of secondary oncogenic alterations^[18]. This dual role in transcriptional regulation and genome maintenance further strengthens the classification of *KMT2C* as a multifunctional tumor suppressor gene.

Collectively, available evidence indicates that *KMT2C* functions as a critical regulator of cellular homeostasis whose disruption contributes to breast cancer development through multiple molecular mechanisms. The recurrent occurrence of loss-of-function variants, combined with experimental observations demonstrating impaired growth regulation following *KMT2C* inactivation, supports the concept that *KMT2C* acts predominantly as a tumor suppressor in breast cancer. Understanding the nature and consequences of *KMT2C* alterations is therefore essential for interpreting their potential impact on disease progression and clinical outcome.

CHAPTER 3

MATERIALS AND METHODS

EXPERIMENTAL METHODOLOGY

3.1 Study Design

The present study was designed as a candidate gene-based molecular genetics study to investigate sequence variants in the *KMT2C* (Lysine Methyltransferase 2C) gene and identify sequence variants in the selected *KMT2C* region and evaluate their potential clinical significance. A total of **9** histopathologically confirmed breast cancer tissue samples and 8 healthy control blood samples were included in the study. Genomic DNA isolated from the collected samples was analyzed using molecular techniques to identify sequence variants within the selected region of the *KMT2C* gene.

3.2 Study Population and Sample Collection

The present study included clinically and histopathologically confirmed breast cancer patients recruited from a local tertiary healthcare centre. Patients belonging to the **35–65 years** age group who fulfilled the eligibility criteria of the study were included after clinical evaluation. Healthy individuals of a comparable age group were recruited as controls.

A small tissue biopsy was obtained from the clinically diagnosed breast lesion during the diagnostic or surgical procedure. The biopsy specimen was sectioned, and a portion of the tissue was immediately transferred into sterile **0.9% normal saline**. The collected tissue samples were stored at **–20°C** until further processing for genomic DNA extraction.

A total of **9 breast cancer tissue samples** and **8 healthy control blood samples** were included in the study. Peripheral blood samples were collected from healthy individuals under sterile conditions and processed according to the laboratory protocol. All samples were assigned unique laboratory identification numbers to ensure proper sample identification and confidentiality throughout the study.

3.3 Chemicals and Reagents

Experimental Stage	Chemical/Reagent	Purpose
DNA Isolation	Tris-HCl	Preparation of tissue lysis and extraction buffers
	Sodium Chloride (NaCl)	Maintains ionic strength during DNA extraction
	EDTA	Chelates divalent metal ions and inhibits DNases
	Sodium Dodecyl Sulfate (SDS)	Cell membrane lysis and protein solubilization
	Proteinase K	Digestion of proteins and histones
	Phenol	Protein denaturation during DNA extraction
	Chloroform	Phase separation during DNA extraction
	Isoamyl Alcohol	Reduces foaming and stabilizes phase separation
	Absolute Ethanol	DNA precipitation
	70% Ethanol	Washing of DNA pellet
	90% Ethanol	Final purification of DNA pellet
	Ammonium Chloride (NH ₄ Cl)	Selective lysis of red blood cells
	Potassium Bicarbonate (KHCO ₃)	Maintains osmotic balance in RBC lysis buffer

	AMQ (Autoclaved Milli-Q) Water	DNA resuspension and reagent preparation
DNA Quantification	Qubit™ dsDNA HS Assay Kit	Fluorometric quantification of genomic DNA
	Thermo Scientific 2× PCR Master Mix	PCR amplification of the KMT2C gene
PCR Amplification	Forward Primer	Specific amplification of the target region
	Reverse Primer	Specific amplification of the target region
	Agarose	Separation of amplified DNA fragments
Agarose Gel Electrophoresis	10× TAE Buffer	Preparation of electrophoresis running buffer
	Ethidium Bromide (EtBr)	Visualization of DNA bands under UV light
PCR Product Cleanup	ExoSAP-IT™	Removal of residual primers and dNTPs prior to sequencing
Sanger Cycle Sequencing	BigDye™ Terminator v3.1 Cycle Sequencing Kit	Fluorescent dye-terminator cycle sequencing
	BigDye™ Terminator v3.1 5× Sequencing Buffer	Maintains optimal reaction conditions during sequencing
	125 mM EDTA	Chelation of metal ions and termination of sequencing reaction
	3 M Sodium Acetate	DNA precipitation
	Absolute Ethanol	Purification of sequencing products

Post-Sequencing Purification	80% Ethanol	Washing of sequencing pellet
	Hi-Di™ Formamide	Denaturation of purified sequencing products before capillary electrophoresis

Table 3.3.1 Chemicals and Reagents Used for DNA Extraction, PCR Amplification, and Sanger Sequencing

3.4 Consumables

Sl. No.	Consumable
A. Genomic DNA Isolation	
1	1.5 mL sterile microcentrifuge tubes
2	2.0 mL sterile microcentrifuge tubes
3	Sterile aerosol-resistant micropipette tips
4	Filter tips
5	Disposable gloves
6	Laboratory masks
B. DNA Quantification (Qubit Analysis)	
7	Qubit assay tubes
8	Sterile aerosol-resistant micropipette tips
C. PCR Amplification	
9	0.2 mL PCR tubes
10	Sterile aerosol-resistant micropipette tips
D. Agarose Gel Electrophoresis	
11	Gel casting tray
12	Electrophoresis comb

13	Electrophoresis tank
E. PCR Product Purification & Sanger Sequencing	
14	Sequencing reaction tubes
15	Parafilm
16	Ice boxes and cooling racks
17	Laboratory tissue wipes

Table 3.4.1 Laboratory Consumables Used for Genomic DNA Isolation, PCR Amplification, and Sanger Sequencing

3.5 Instruments and Equipment

The instruments used in the present study are listed in **Table 3.5.1**.

Sl. No.	Instrument	Application
1	Qubit™ Fluorometer	DNA quantification
2	Thermal Cycler	PCR amplification
3	ABI 3500 Genetic Analyzer	Capillary electrophoresis and Sanger sequencing
4	Refrigerated Microcentrifuge	DNA extraction
5	Bench-top Centrifuge	Routine centrifugation
6	Vortex Mixer	Mixing reagents
8	Water Bath	DNA dissolution and incubation
9	Gel Electrophoresis Unit	Separation of PCR products
10	UV Transilluminator/Gel Documentation System	Visualization of DNA bands

11	Analytical Balance	Preparation of reagents
12	Microwave Oven	Dissolution of agarose
13	Adjustable Micropipettes (P2, P10, P20, P200, P1000)	Accurate liquid dispensing
14	Refrigerator (4°C)	Reagent storage
15	Deep Freezer (−20°C)	DNA and reagent storage

Table 3.5.1 Instruments and Equipment Used During Molecular Analysis of Breast Cancer Samples

3.6 Preparation of Buffers and Reagents

Reagent	Weight	Stock Solution Concentration	pH Adjusted To	Final Volume	Working Standard Solution Concentration	Volume of Working Standard Used
Tris-HCl	7.88 g	1 M	7.8–8.0	50 mL	10 mM	0.5 mL
Sodium Chloride (NaCl)	14.61 g	5 M	—	50 mL	10 mM	0.1 mL
EDTA	9.3 g	0.5 M	8.0	50 mL	25 mM	2.5 mL
Sodium Dodecyl Sulfate (SDS)	10 g	20% (w/v)	—	50 mL	0.5%	25 ul

Table 3.6 Reagents Used for the Preparation of Stock Solutions and Working Buffers

3.6.1 Preparation of Soft Tissue Lysis Buffer

The soft tissue lysis buffer was prepared using the stock solutions listed above.

Component	Final Concentration
Tris-HCl	10 mM
NaCl	10 mM
EDTA	25 mM
SDS	0.5%
Proteinase K	0.1 mg/mL

Table 3.6.1 Reagents for Soft Tissue Lysis Buffer

Composition of Soft Tissue Lysis Buffer

A working soft tissue lysis buffer was prepared by mixing **0.5 mL of 1 M Tris-HCl**, **0.1 mL of 5 M NaCl**, and **2.5 mL of 0.5 M EDTA** in sterile deionized water.

1. **Twenty percent SDS** was added to obtain a final concentration of **0.5%**, followed by the addition of **Proteinase K** immediately before use to achieve a final concentration of **0.1 mg/mL**.
2. The volume was made up to **50 mL** using sterile deionized water, and the buffer was mixed thoroughly to obtain a homogeneous solution before use.

3.6.2 Preparation of Buffers for Whole Blood DNA Isolation

Two buffers were prepared for genomic DNA isolation from whole blood samples: **Red Blood Cell (RBC) Lysis Buffer** and **White Blood Cell (WBC) Lysis Buffer**.

3.6.2.1 Red Blood Cell (RBC) Lysis Buffer

Component	Final Concentration
Ammonium chloride (NH ₄ Cl)	155 mM
Potassium bicarbonate (KHCO ₃)	10 mM
EDTA	0.1 mM
pH	7.2–7.4

Table 3.6.2.1 Reagents For RBC Lysis Buffer

Preparation

1. The RBC lysis buffer was prepared according to the composition presented in **Table 3.6.2.1**, and the pH was adjusted to **7.2–7.4**.
2. The final volume was adjusted with sterile deionized water, and the prepared buffer was mixed thoroughly and stored at **4°C** until use.

3.6.2.2 White Blood Cell (WBC) Lysis Buffer

Component	Final Concentration
Tris-HCl	10 mM
NaCl	100 mM
EDTA	25 mM

SDS	0.5%
pH	8.0

Table 3.6.2.2 Reagents For WBC Lysis Buffer

Preparation

1. The WBC lysis buffer was prepared according to the composition shown in Table 3.6.3.1.
2. The prepared buffer was mixed thoroughly until a homogeneous solution was obtained and stored at room temperature until use.

3.7 Genomic DNA Isolation from Breast Tissue Samples

Breast tissue samples were obtained from **clinically and histopathologically confirmed breast cancer patients** recruited from a local tertiary healthcare centre. A small portion of the breast tissue biopsy was transferred into sterile **0.9% normal saline** immediately after collection and stored at **-20°C** until genomic DNA extraction.

Principle

Genomic DNA was isolated from the collected breast tissue samples using the **Phenol–Chloroform–Isoamyl Alcohol (PCI)** extraction method. The method is based on selective separation of nucleic acids from proteins and other cellular components through organic extraction, followed by ethanol precipitation to obtain purified genomic DNA suitable for downstream PCR amplification and Sanger sequencing.

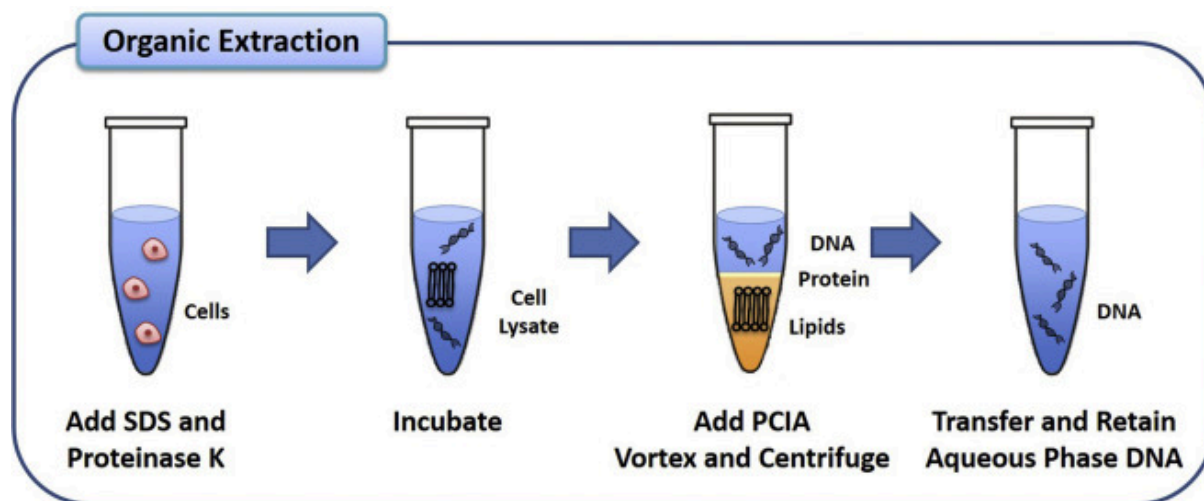


Figure 3.7.1. Workflow of genomic DNA isolation using the Phenol–Chloroform–Isoamyl Alcohol (PCI) extraction method.

Application

The Phenol–Chloroform–Isoamyl Alcohol (PCI) extraction method is widely used for isolating high-quality genomic DNA suitable for PCR amplification, DNA sequencing, genotyping, mutation analysis, and other molecular biology applications.

Step 1: Sample Thawing

- 1.1 Breast tissue samples stored at -20°C were removed from the freezer.
- 1.2 The samples were placed **on ice** and allowed to thaw completely.
- 1.3 The thawed samples were gently mixed before further processing.

Step 2: Transfer of Tissue Sample

- 2.1 The thawed tissue suspension was transferred into sterile **2 mL microcentrifuge (MCT) tubes** using cut pipette tips.
- 2.2 The Falcon tube was briefly centrifuged to collect any remaining tissue suspension.
- 2.3 The remaining suspension was transferred into the corresponding MCT tube.

Step 3: Initial Centrifugation

- 3.1 The samples were centrifuged at **13,000 rpm for 3 minutes**.
- 3.2 The supernatant was carefully discarded without disturbing the tissue pellet.
- 3.3 The tissue pellet was retained for subsequent DNA extraction.

Step 4: Addition of Soft Tissue Lysis Buffer

- 4.1 Soft tissue lysis buffer was added to each tissue pellet according to the optimized extraction protocol.
- 4.2 Tissue pellets belonging to the same sample were pooled into a single microcentrifuge tube.
- 4.3 The suspension was mixed thoroughly until a homogeneous mixture was obtained.

Step 5: Addition of SDS and Proteinase K

- 5.1 **20% Sodium Dodecyl Sulfate (SDS)** was added to obtain the required final concentration in the lysis mixture.
- 5.2 **Proteinase K** was added according to the tissue quantity to facilitate enzymatic digestion.
- 5.3 The contents were mixed gently to ensure uniform distribution of the reagents.

Step 6: Overnight Digestion

- 6.1 The tubes were incubated under **continuous rotation at 37°C overnight**.

Step 7: Phenol Extraction and Recovery of Aqueous Phase

- 7.1 An **equal volume of phenol** was added to the lysed sample, followed by gentle mixing for **10 minutes**.
- 7.2 The samples were centrifuged at **10,000 rpm for 10 minutes**.

- 7.3 The upper aqueous phase was carefully transferred into a fresh sterile **2 mL microcentrifuge tube** using cut pipette tips.

Step 8: Phenol–Chloroform–Isoamyl Alcohol Extraction

- 8.1 An **equal volume of Phenol:Chloroform:Isoamyl Alcohol (25:24:1)** was added to the recovered aqueous phase.
- 8.2 The mixture was gently mixed for **10 minutes** and centrifuged at **10,000 rpm for 10 minutes**.
- 8.3 The upper aqueous phase was carefully transferred into a fresh sterile microcentrifuge tube.

Step 9: Chloroform–Isoamyl Alcohol Extraction and Recovery of Purified DNA Solution

- 9.1 An **equal volume of Chloroform:Isoamyl Alcohol (24:1)** was added to the recovered aqueous phase.
- 9.2 The mixture was gently mixed for **10 minutes** and centrifuged at **10,000 rpm for 10 minutes**.
- 9.3 The upper aqueous phase containing purified genomic DNA was carefully transferred into a fresh sterile microcentrifuge tube.

Step 10: Ethanol Precipitation of Genomic DNA

- 10.1 An **1ml of chilled absolute ethanol** was added to the purified DNA solution and mixed gently by inversion to complete precipitation
- 10.2 The tubes were centrifuged at **13,000 rpm for 15 minutes** to pellet the genomic DNA.

Step 11: Ethanol Washing of Genomic DNA Pellet

- 11.1 The DNA pellet was washed sequentially with **500 µL of 70% ethanol** followed by **500 µL of 90% ethanol**.
- 11.2 After each wash, the samples were mixed gently and centrifuged at **14,000 rpm for 7 minutes**.

- 11.3 The supernatant was carefully discarded after each centrifugation without disturbing the DNA pellet.

Step 12: Drying and Reconstitution of the DNA Pellet

- 12.1 The residual ethanol was completely discarded, and the DNA pellet was **air-dried** until all traces of ethanol had evaporated.
- 12.2 The dried DNA pellet was reconstituted in **50 µL of warm (52°C) autoclaved Milli-Q (AMQ) water** and mixed gently until completely dissolved.

Step 13: Storage of Genomic DNA

- 13.1 The reconstituted genomic DNA samples were subsequently stored at **−20°C** until further use for DNA quantification, PCR amplification, and Sanger sequencing.

3.8 Genomic DNA Isolation from Whole Blood (Control Samples)

Genomic DNA was isolated from peripheral whole blood samples collected from healthy control individuals. Prior to DNA extraction, red blood cells were selectively lysed to obtain a purified leukocyte pellet, which was subsequently processed using the Phenol–Chloroform–Isoamyl Alcohol (PCI) extraction method.

Principle

Genomic DNA isolation from whole blood is based on the selective removal of mature erythrocytes, which are anucleate, while preserving nucleated leukocytes as the source of genomic DNA. The purified leukocyte pellet was processed by the PCI extraction method to obtain high-quality genomic DNA suitable for downstream molecular analyses.



Figure 3.8.1. Laboratory setup used for genomic DNA isolation by the Phenol–Chloroform–Isoamyl Alcohol (PCI) extraction method.

Application

Genomic DNA isolated from peripheral blood is routinely used for PCR, DNA sequencing, SNP genotyping, mutation screening, and molecular diagnostic studies.

Step 1: Collection of Whole Blood Sample

- 1.1 Approximately **500 μ L** of peripheral whole blood was transferred into a sterile **2 mL microcentrifuge tube**.
- 1.2 The sample was gently inverted to obtain a uniform suspension before further processing.

Step 2: Red Blood Cell (RBC) Lysis

- 2.1 **Three volumes of RBC lysis buffer** were added to the blood sample, followed by gentle inversion to ensure thorough mixing.
- 2.2 The sample was incubated for **10 minutes**, centrifuged at **3000 rpm for 5 minutes**, and the supernatant was carefully discarded without disturbing the leukocyte pellet.

Step 3: Repeated RBC Lysis

- 3.1 The leukocyte pellet was resuspended in RBC lysis buffer and centrifuged again at **3000 rpm for 5 minutes**.
- 3.2 The washing procedure was repeated until a distinct **white leukocyte pellet** was obtained.

Step 4: Addition of White Blood Cell (WBC) Lysis Buffer

- 4.1 The leukocyte pellet was resuspended in **500 µL of WBC lysis buffer** until a homogeneous suspension was obtained.

Step 5: Genomic DNA Isolation

- 5.1 Following WBC lysis, genomic DNA isolation was performed using the **Phenol–Chloroform–Isoamyl Alcohol (PCI) extraction protocol** described in **Section 3.7 (Steps 5–13)**, including **Proteinase K digestion, organic extraction, ethanol precipitation, ethanol washing, DNA reconstitution, and storage**.

3.9 DNA Quantification Using Qubit™ Fluorometer

Following genomic DNA isolation, DNA concentration was determined using the **Qubit™ Fluorometer (Thermo Fisher Scientific, USA)** in combination with the **Qubit™ dsDNA High Sensitivity (HS) Assay Kit** according to the manufacturer's instructions.

Principle

The Qubit™ Fluorometer quantifies genomic DNA based on fluorescence generated by a dsDNA-specific fluorescent dye. The dye selectively binds to double-stranded DNA and emits fluorescence proportional to the DNA concentration. The measured fluorescence is converted into DNA concentration (ng/µL), enabling accurate quantification of genomic DNA for subsequent PCR amplification and Sanger sequencing.

Application

The Qubit™ 3.0 Fluorometer is a fluorescence-based instrument designed for accurate quantification of nucleic acids and proteins. Using assay-specific fluorescent dyes, it can quantify dsDNA, ssDNA, RNA, miRNA, oligonucleotides, and proteins with high sensitivity and specificity. It is widely used in PCR, Sanger sequencing, NGS, library preparation, cloning, and molecular diagnostic applications. In this study, the Qubit™ dsDNA High Sensitivity (HS) Assay Kit was employed to quantify genomic DNA before PCR amplification.



Figure 3.9.1. Qubit™ 3.0 Fluorometer used for fluorometric quantification of genomic DNA.

Step 1: Preparation of Qubit Working Solution

1.1 The **Qubit™ dsDNA HS Assay Kit** was allowed to equilibrate to room temperature prior to use.

A fresh working solution was prepared according to the composition shown in **Table 3.9.1**.

Component	Volume
Qubit HS Buffer	198 μL
Qubit HS Reagent (Dye)	1 μL
Total Working Solution	199 μL

Table 3.9.1 Composition of the Qubit™ dsDNA HS Working Solution Used for DNA Quantification

Step 2: Sample Preparation

2.11 μL of genomic DNA was combined with **199 μL** of freshly prepared Qubit™ working solution according to **Table 3.9.2**.

2.2 The assay mixture was gently mixed by pipetting and incubated at **room temperature for approximately 5 minutes in the dark**.

The composition of the assay mixture is presented in **Table 3.9.2**:

Component	Volume
Working Solution	199 μL
DNA Sample	1 μL
Final Volume	200 μL

Table 3.9.2 Composition of the Qubit™ DNA Assay Mixture Used for DNA Quantification

Step 3: DNA Quantification and Selection of Samples

3.1 The prepared assay tubes were placed in the **Qubit™ Fluorometer**, and DNA concentration was measured according to the manufacturer's instructions.

3.2 DNA concentration was recorded in **ng/ μL** .

3.3 DNA samples exhibiting adequate concentration and acceptable quality were selected for subsequent PCR amplification.

3.10 Primer Design and Primer Preparation

Gene-specific primers targeting the selected region of the **KMT2C** gene were used for PCR amplification. Primer characteristics, including melting temperature (T_m), primer length, GC content, amplification specificity, and expected amplicon size, were evaluated prior to synthesis. The primer pair generated an expected amplicon of **378 bp**.

Principle

Primer design is based on the complementary base-pairing of short synthetic oligonucleotides to specific target sequences flanking the region of interest in the **KMT2C** gene. Properly designed primers with appropriate melting temperature (T_m), GC content, and sequence specificity ensure accurate and efficient amplification of the desired 378 bp target fragment during PCR. The primers were subsequently prepared at the required working concentration for reproducible PCR amplification.

Primer Design Software

The primer pair targeting the selected region of the **KMT2C** gene was designed using the **Primer3** web server (<https://primer3.ut.ee/>). The target **KMT2C** reference sequence used for primer design was retrieved from the NCBI Gene database prior to primer design. Primer specificity was subsequently verified using the **NCBI Primer-BLAST** tool to ensure amplification of only the intended target region within the human genome. During primer design, parameters including primer length, melting temperature (T_m), GC content, self-complementarity, hairpin formation, primer-dimer formation, and expected amplicon size were optimized to obtain specific and efficient amplification.

Application

Primer design software such as Primer3 and Primer-BLAST is extensively used in molecular biology to design highly specific primers for PCR amplification, DNA sequencing, mutation analysis, genotyping, and molecular diagnostics.

Primer Design Criteria

The primer design parameters used for amplification of the target **KMT2C** gene region are presented in **Table 3.10.1**.

Primer Sequences Used

Parameter	Selection Criteria
Primer Length	18–25 nucleotides
Melting Temperature (T _m)	Approximately 58–60°C
GC Content	Approximately 40–60%
Secondary Structures	Minimized
Primer Dimer Formation	Avoided
Hairpin Formation	Avoided
Amplification Specificity	High
Expected Amplicon Size	378 bp

Table 3.10.1 Primer Design Parameters Considered for Amplification of the KMT2C Gene

The forward and reverse primers used for amplification of the selected **KMT2C** gene region are presented in **Table 3.10.2**.

Primer	Sequence (5' → 3')
Forward Primer	GCTGACGAATCATGGCTTGC
Reverse Primer	GTTTGCCTCCTTTTCATAGCAC

Table 3.10.2 Primer Sequences Used for Amplification of the KMT2C Gene

Primer Characteristics

Parameter	Forward Primer	Reverse Primer
Length	20 bp	21 bp
Melting Temperature (T _m)	59.3°C	57.9°C
Stock Concentration	100 pmol/μL	100 pmol/μL
Working Concentration	05 pmol/μL	05 pmol/μL

Table 3.10.3 Primer Characteristics

Step 1: Preparation of Primer Stock Solution

The lyophilized primers supplied by the manufacturer were reconstituted according to the manufacturer's instructions to obtain a **100 pmol/μL stock solution**.

Step 2: Preparation of Working Primer Solution

A **5 pmol/μL working primer solution** was prepared by diluting the **100 pmol/μL primer stock** in a **1:20 ratio** using autoclaved Milli-Q (AMQ) water. The composition of the working primer solution is presented in **Table 3.10.4**.

Component	Volume
Primer Stock (100 pmol/μL)	5 μL
Autoclaved Milli-Q (AMQ) Water	95 μL
Final Volume	100 μL

Table 3.10.4 Preparation of Primer Working Solution

Step 3: Storage of Primers

Both stock and working primer solutions were stored at -20°C until use, while repeated freeze–thaw cycles were avoided to maintain primer stability.

Parameter	Forward Primer	Reverse Primer
Sequence (5'→3')	GCTGACGAATCATGGCTTGC	GTTTGCCTCCTTTTCATAGCAC
Length (bp)	20	21
T _m (°C)	59.3	57.9
Stock Concentration	100 pmol/μL	100 pmol/μL
Working Concentration	5 pmol/μL	5 pmol/μL

Table 3.10.5 Primer Specifications

3.11 PCR Amplification of the KMT2C Gene

PCR Reaction Mixture

PCR amplification of the target KMT2C gene region was performed using gene-specific primers and Thermo Scientific 2× PCR Master Mix. All reaction components were prepared on ice to minimize non-specific primer annealing and preserve enzyme activity. A master mix containing all reaction components except the DNA template was prepared to improve reproducibility.

Principle

Polymerase Chain Reaction (PCR) is based on the enzymatic amplification of a specific DNA sequence through repeated cycles of denaturation, primer annealing, and extension. Gene-specific primers selectively bind to complementary regions flanking the target KMT2C gene sequence, while a thermostable DNA polymerase synthesizes new DNA strands during each cycle. Repeated amplification results in exponential accumulation of the target DNA

fragment, producing sufficient quantities of the expected 378 bp amplicon for agarose gel electrophoresis and subsequent Sanger sequencing.

Application

PCR is widely employed in molecular biology for gene amplification, mutation detection, pathogen identification, genotyping, molecular diagnostics, forensic analysis, and DNA sequencing applications.

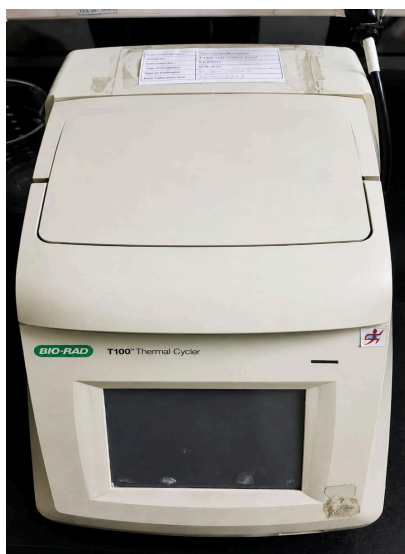


Figure 3.11.1. Bio-Rad T100™ Thermal Cycler used for PCR amplification and Sanger cycle sequencing.

Component	Volume (μL)
Thermo Scientific 2× PCR Master Mix	5.0
Forward Primer (10 pmol/μL)	0.2
Reverse Primer (10 pmol/μL)	0.2
Nuclease-free Water	3.6
Genomic DNA Template	1.0
Total Volume	10.0

Table 3.11.1 Composition of PCR Reaction Mixture (10 μL Reaction Volume)

Step 1: Preparation of PCR Master Mix

- 1.1 PCR reagents were thawed on ice and mixed gently before use.
- 1.2 A master mix containing Thermo Scientific 2× PCR Master Mix, forward primer, reverse primer, and nuclease-free water was prepared and aliquoted into sterile PCR tubes.
- 1.3 Finally, 1 µL of genomic DNA was added individually to each reaction tube.

Step 2: PCR Amplification

- 2.1 The PCR tubes were placed in a thermal cycler programmed with the following optimized cycling conditions.

Step	Temperature	Time	Number of Cycles
Initial Denaturation	95°C	5 min	1
Denaturation	95°C	35 sec	35
Annealing	57°C	40 sec	35
Extension	72°C	45 sec	35
Final Extension	72°C	5 min	1
Hold	4°C	Until retrieval	—

Table 3.11.2 Optimized Thermal Cycling Conditions Used for PCR Amplification of the KMT2C Gene

3.12 Agarose Gel Electrophoresis

PCR products were analyzed by **2% agarose gel electrophoresis** prepared in **1× TAE buffer** and stained with **ethidium bromide (EtBr)**. Electrophoresis was performed to confirm amplification of the expected **378 bp KMT2C amplicon**.

Principle

Agarose gel electrophoresis is based on the migration of negatively charged DNA molecules through a porous agarose matrix under the influence of an electric field. DNA fragments are separated according to their size, with smaller fragments migrating faster than larger ones. Ethidium bromide intercalates between the DNA base pairs and fluoresces under ultraviolet (UV) light, enabling visualization of the separated DNA bands. In this study, agarose gel electrophoresis was used to verify successful PCR amplification of the KMT2C gene by confirming the presence of the expected 378 bp amplicon.

Application

Agarose gel electrophoresis is routinely used to evaluate DNA quality, confirm PCR amplification, estimate DNA fragment size, verify restriction enzyme digestion, and assess nucleic acids prior to downstream sequencing applications.

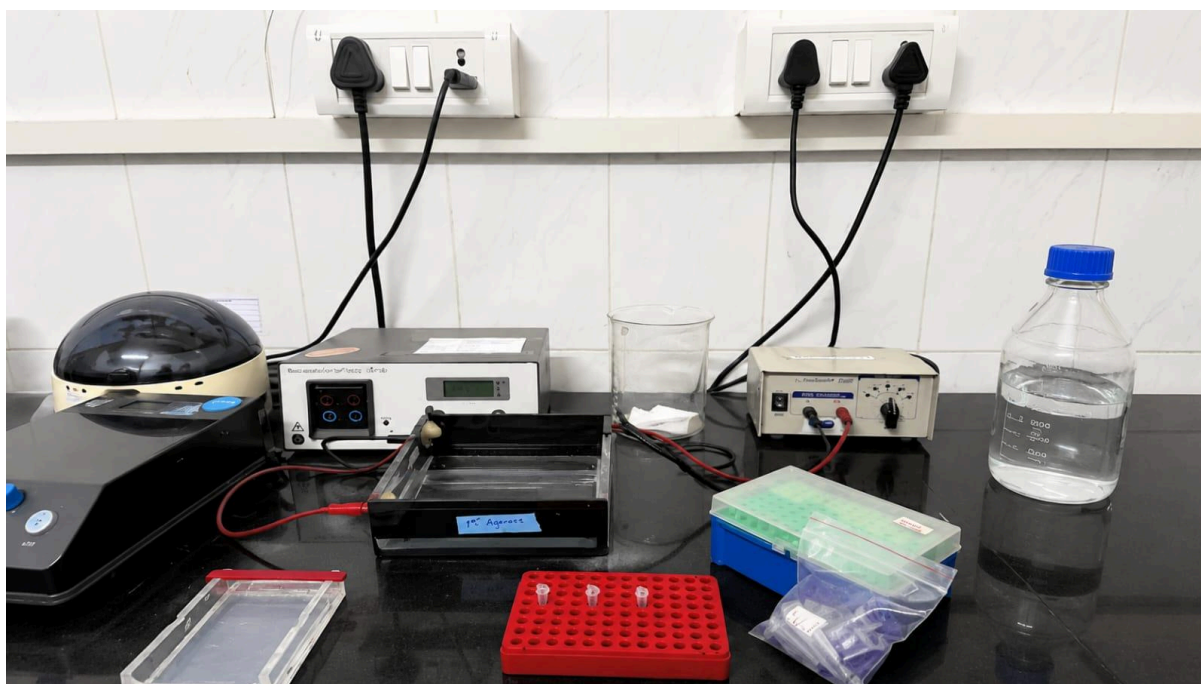


Figure 3.12.1. Laboratory setup used for agarose gel electrophoresis.

Step 1: Preparation of Agarose Gel

1.10.9 g of agarose was dissolved in 45 mL of 1× TAE buffer by heating in a microwave oven until completely dissolved.

1.2 The molten agarose was cooled to approximately **55–60°C**, followed by the addition of **3 µL ethidium bromide (EtBr)** and mixed thoroughly.

1.3 The agarose solution was poured into a gel casting tray fitted with a comb to form sample wells.

1.4 The gel was allowed to solidify completely at room temperature.

Step 2: Preparation of the Electrophoresis Chamber

2.1 The solidified gel was transferred to the electrophoresis chamber and completely submerged in **1× TAE buffer**.

Step 3: Loading of Samples

3.1 **2 µL of 100bp DNA ladder** was loaded into the first well, followed by **3 µL of each PCR product** into the remaining wells.

Step 4: Electrophoresis

4.1 The gel was electrophoresed at **150 V for approximately 20 minutes**.

Step 5: Visualization

5.1 Following electrophoresis, the gel was visualized under a **UV transilluminator**, and images were documented for analysis.

5.2 Successful PCR amplification was confirmed by the presence of the expected **378 bp DNA band**.

3.13 Sanger Cycle Sequencing

Purified PCR products were subjected to Sanger cycle sequencing using the **BigDye™ Terminator v3.1 Cycle Sequencing Kit (Applied Biosystems, Thermo Fisher Scientific, USA)**. Cycle sequencing was performed using the **forward primer**, followed by capillary electrophoresis for sequence analysis.

Principle

Sanger cycle sequencing is based on the selective incorporation of fluorescently labeled dideoxynucleotide triphosphates (ddNTPs) during DNA synthesis, resulting in chain termination at specific nucleotide positions. A single forward primer initiates DNA synthesis in the presence of DNA polymerase, deoxynucleotide triphosphates (dNTPs), and dye-labeled ddNTPs, generating extension products of varying lengths. The resulting fragments are subsequently separated by capillary electrophoresis, and the fluorescent signals emitted by the terminal ddNTPs are detected to determine the nucleotide sequence of the target DNA. In this study, Sanger cycle sequencing was employed to analyze the amplified KMT2C gene region for the identification of sequence variants.

Application

Sanger sequencing is considered the gold standard for validation of genetic variants and is widely used for mutation detection, SNP analysis, clinical genetic testing, confirmation of next-generation sequencing results, and molecular diagnostics.

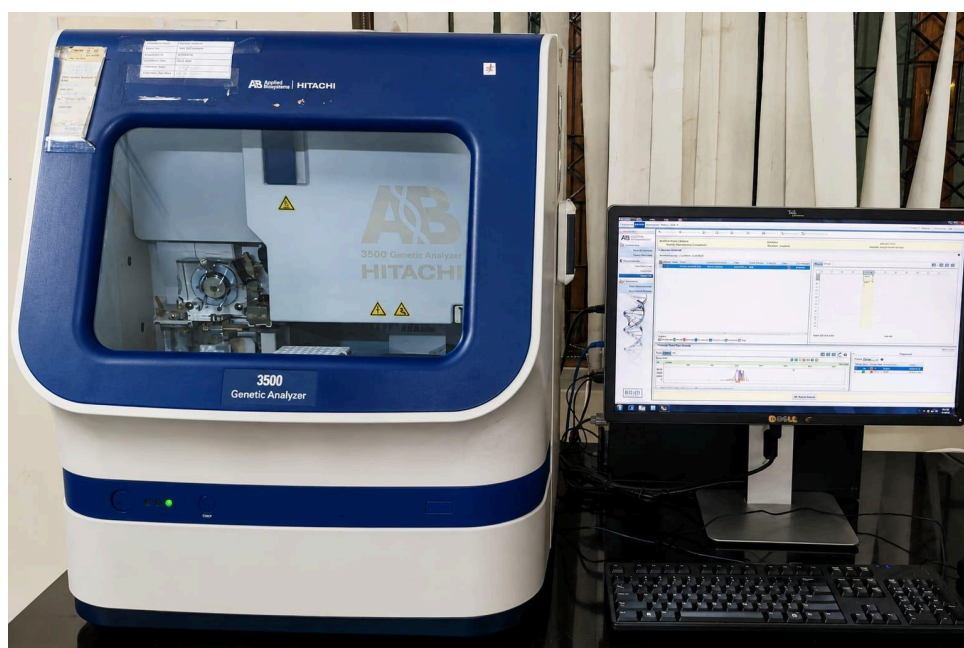


Figure 3.15.1. ABI 3500 Genetic Analyzer used for capillary electrophoresis and DNA sequence analysis.

3.13.1 Pre-Cleanup of PCR Products (ExoSAP-IT™)

Procedure

1. **2 µL** of PCR product was mixed with **0.8 µL ExoSAP-IT™** reagent and briefly centrifuged.
2. The reaction mixture was subjected to the ExoSAP-IT™ thermal cycling program shown in **Table 3.13.1**.

Stage	Temperature	Time
Enzymatic digestion	37°C	4 min
Heat inactivation	80°C	1 min
Hold	4°C	∞

Table 3.13.1 Thermal Cycling Conditions Used for ExoSAP-IT™ Enzymatic Cleanup

3. The purified PCR products were used directly for sequence PCR.

3.13.2 Preparation of Sequence PCR Reaction

Sequence PCR reactions were prepared in a final reaction volume of **10 μ L**.

Component	Volume (μ L)
BigDye™ Terminator v3.1 Ready Reaction Mix	0.5
5× Sequencing Buffer	2.0
Nuclease-free Water	4.5
Forward Primer (2.5 pmol)	1.0
Purified PCR Product (Template DNA)	2.0
Total Volume	10.0

Table 3.13.2 Composition of Sequence PCR Reaction Mixture for BigDye™ Cycle Sequencing

3.13.3 Cycle Sequencing Conditions

The sequencing reaction was amplified using the optimized cycle sequencing conditions presented in **Table 3.13.3**.

Step	Temperature	Time	Cycles
Initial Denaturation	96°C	1 min	1
Denaturation	96°C	10 sec	25
Annealing	50°C	5 sec	25
Extension	60°C	4 min	25
Hold	4°C	∞	—

Table 3.13.3 Optimized Thermal Cycling Conditions for BigDye™ Cycle Sequencing

3.14 Post-Sequencing Purification

Following cycle sequencing, the extension products were purified to remove excess fluorescent dye terminators and reaction components before capillary electrophoresis.

3.14.1 Preparation of Cleanup Solutions

Two cleanup solutions (Master Mix I and Master Mix II) were prepared according to the compositions presented in Tables 3.14.1.1 and 3.14.1.2.

Master Mix I

Component	Volume per Reaction
Nuclease-free Water	10 μ L x n
125 mM EDTA	2 μ L x n

Table 3.14.1.1 Composition of Master Mix I for Post-Sequencing Purification

Master Mix II

Component	Volume per Reaction
3 M Sodium Acetate	2 μ L x n
Absolute Ethanol	50 μ L x n

Table 3.14.1.2 Composition of Master Mix II for Post-Sequencing Purification

n = (number of reactions)

3.14.2 Purification Procedure

1. 10 μ L of sequence PCR product was transferred into a sterile 1.5 mL microcentrifuge tube, followed by the addition of 12 μ L Master Mix I and 52 μ L Master Mix II.

2. The reaction mixture was mixed gently, incubated in the dark for **15 minutes**, and centrifuged at **13,000 rpm for 20 minutes**.
3. The supernatant was discarded, and the pellet was washed with **250 µL of 80% ethanol**, followed by centrifugation at **13,000 rpm for 10 minutes**.
4. The ethanol was discarded, the pellet was air-dried for **1–1.5 hours**, and reconstituted in **9 µL Hi-Di™ Formamide** before brief centrifugation.

3.15 Capillary Electrophoresis Using ABI 3500 Genetic Analyzer

The purified sequencing products were analyzed using the **ABI 3500 Genetic Analyzer** (Applied Biosystems, Thermo Fisher Scientific, USA).

Principle

The ABI 3500 Genetic Analyzer performs DNA sequence analysis by capillary electrophoresis. Fluorescently labeled DNA fragments generated during Sanger sequencing are separated according to fragment length within polymer-filled capillaries. A laser detects the fluorescent dye attached to each terminal nucleotide, and the emitted fluorescence is converted into electropherograms representing the nucleotide sequence.

Application

The ABI 3500 Genetic Analyzer is widely used for Sanger sequencing, fragment analysis, STR genotyping, mutation detection, microsatellite analysis, and clinical molecular diagnostics.

Procedure

1. **10 µL** of purified sequencing product was dispensed into the designated wells of a 96-well sequencing plate.
2. The sequencing plate was loaded into the ABI 3500 Genetic Analyzer, and electrophoresis was performed using the StdSeq Assay with POP-7 polymer.

3. Chromatogram files (**.ab1**) were generated automatically upon completion of the sequencing run.

3.16 Sequence Analysis Using CodonCode

The chromatogram files generated by the ABI 3500 Genetic Analyzer were analyzed using **CodonCode** software for identification of sequence variants in the *KMT2C* gene.

Software Features

CodonCode software provides visualization of sequencing chromatograms, sequence assembly, alignment with the reference sequence, manual verification of nucleotide calls, detection of sequence variants, and editing of low-quality sequence regions. These features facilitate accurate identification and confirmation of genetic variants obtained through Sanger sequencing.

Procedure

1. The generated **.ab1 chromatogram files** were imported into **CodonCode** software.
2. Chromatograms were assessed for sequence quality, and low-quality regions were trimmed where necessary.
3. The obtained sequences were aligned with the reference *KMT2C* sequence to identify nucleotide variations.
4. Identified variants were verified by manual inspection of the chromatograms and recorded for subsequent analysis.

3.17 Mutation Analysis

Mutation analysis was performed by comparing the obtained *KMT2C* sequences with the corresponding reference sequence. Sequence variations identified during chromatogram analysis were recorded and interpreted.

Application

Mutation analysis is widely used to identify disease-associated genetic variants, investigate genotype–phenotype relationships, support molecular diagnosis, evaluate prognostic biomarkers, and facilitate precision medicine approaches in cancer research.

Step 1: Variant Identification

- 1.1 Nucleotide changes observed in patient samples were compared with the reference sequence.
- 1.2 Sequence variants were documented.

Step 2: Confirmation of Variants

- 2.1 Each identified variant was verified by careful examination of the sequencing chromatogram to ensure that the observed nucleotide change represented a true sequence variation rather than a sequencing artifact.

Step 3: Documentation

- 3.1 Confirmed sequence variants were recorded for each sample.
- 3.2 Mutation data were subsequently used for interpretation and correlation with the clinical outcomes of breast cancer patients.

CHAPTER 4

RESULTS

RESULTS:

4.1 PCR Amplification of KMT2C Gene

A total of 12 breast cancer samples were collected for the identification and characterization of variants in the *KMT2C* gene. PCR amplification of the target region was performed using gene-specific primers. One sample failed to produce amplification and was excluded from further analysis. Among the remaining 11 PCR-positive samples, three samples exhibited faint amplification bands on agarose gel electrophoresis and were therefore excluded from sequencing analysis. Consequently, 8 samples with clear PCR products were selected for Sanger sequencing.

S.No	Sample code	Variant Status
1	GFTNBC060	Variant detected
2	GFTNBC036	No variant detected
3	GFTNBC077	No variant detected
4	GFTNBC082	No variant detected
5	GFTNBC079	No variant detected
6	GFTNBC080	No variant detected
7	GFTNBC090	No variant detected
8	GFTNBC075	No variant detected

Table 4.1.1 Mutation screening results of the KMT2C gene in breast cancer samples following PCR amplification and Sanger sequencing.

Control:

Control samples were DNA samples obtained from individuals without breast cancer and were used as reference standards for comparison with breast cancer samples during mutation analysis. These control samples were used as reference materials during the molecular

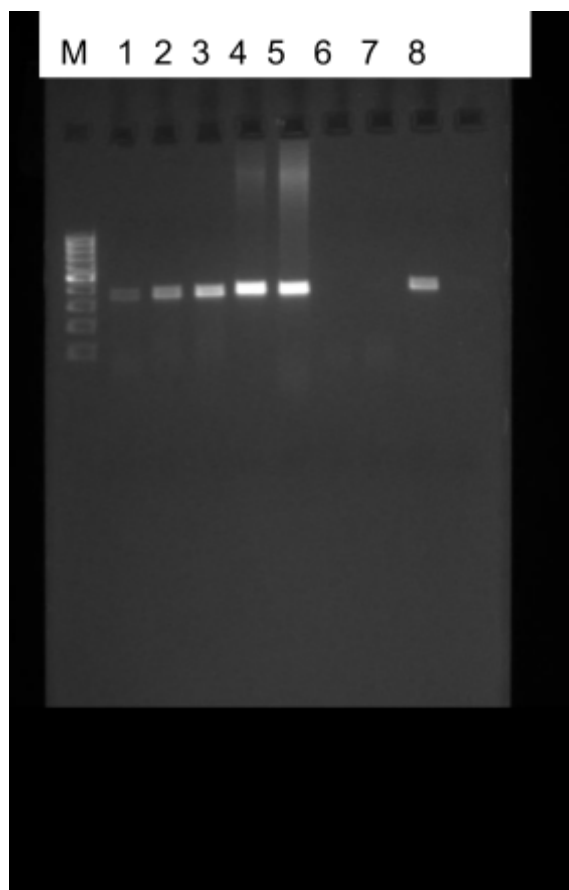
analysis to ensure the accuracy, reliability, and reproducibility of PCR amplification and Sanger sequencing results. The inclusion of control samples helped validate the experimental procedures and served as a benchmark for comparison with the breast cancer samples analyzed for *KMT2C* gene mutations.

S.No	Control Sample
1	2918B
2	2934B
3	2913A
4	2971C
5	2947B
6	GF001
7	GF002
8	GF003

Table 4.1.2 List of control samples included during PCR amplification and Sanger sequencing for quality assurance.

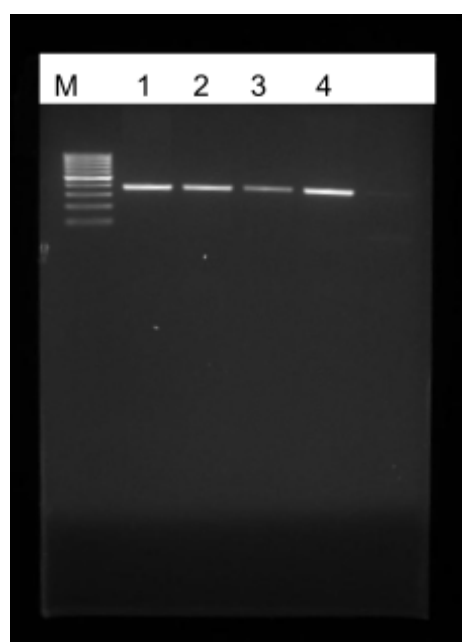
4.2 Agarose Gel Electrophoresis

PCR products were resolved on an agarose gel to verify the specificity and integrity of amplification. Most samples showed a single, sharp band at the expected product size, while samples without amplification or with poor-quality bands were excluded from further analysis. The absence of amplification in the negative control confirmed that no contamination occurred during PCR.



M-Ladder
1-GFTNBC082
2-GFTNBC077
3-GFTNBC078
4-GFTNBC090
5-GFTNBC060
6-GFTNBC011
7-GFTNBC036
8-Negative Control

Fig 4.2.1 Agarose gel electrophoresis showing PCR amplification of the KMT2C gene. The gel was visualized and documented using a Gel Doc imaging system. Lane M: 100 bp DNA ladder; Lanes 1–7: breast cancer samples (GFTNBC082, GFTNBC077, GFTNBC078, GFTNBC090, GFTNBC060, GFTNBC011, and GFTNBC036); Lane 8: Negative control.



M-Ladder
1-GFTNBC087
2-GFTNBC062
3-GFTNBC075
4-Negative Control

Fig 4.2.2 Agarose gel electrophoresis showing PCR amplification of the *KMT2C* gene. The gel image was visualized and documented using a Gel Documentation (Gel Doc) System. Lane M: 100 bp DNA ladder; Lanes 1–3: breast cancer samples (GFTNBC087, GFTNBC062, and GFTNBC075); Lane 4: Negative control.

4.3 Sanger Sequencing Analysis

The PCR-amplified products were subjected to Sanger sequencing for mutation analysis of the *KMT2C* gene. The obtained sequencing chromatograms were analyzed using CodonCode Aligner software by aligning the sample sequences with the reference *KMT2C* gene sequence. The software facilitated sequence assembly, chromatogram quality assessment, base calling, and identification of nucleotide variations. The detected variants were further interpreted to determine the presence or absence of mutations in the analyzed samples. Chromatograms were carefully examined to distinguish homozygous wild-type, heterozygous, and homozygous variant genotypes.

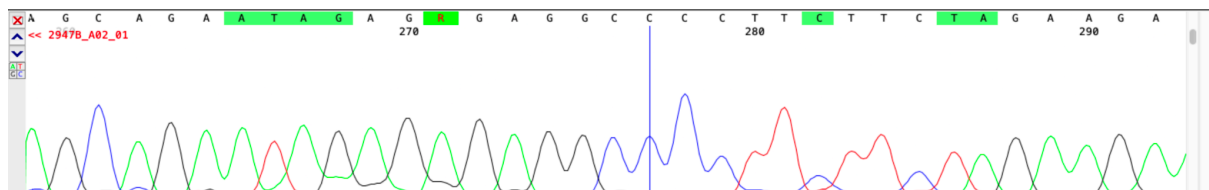


Fig 4.3.1 Representative Sanger sequencing chromatogram of a normal (wild-type) *KMT2C* gene sequence analyzed using CodonCode Aligner software. The chromatogram shows a clear nucleotide sequence with no detectable mutations.

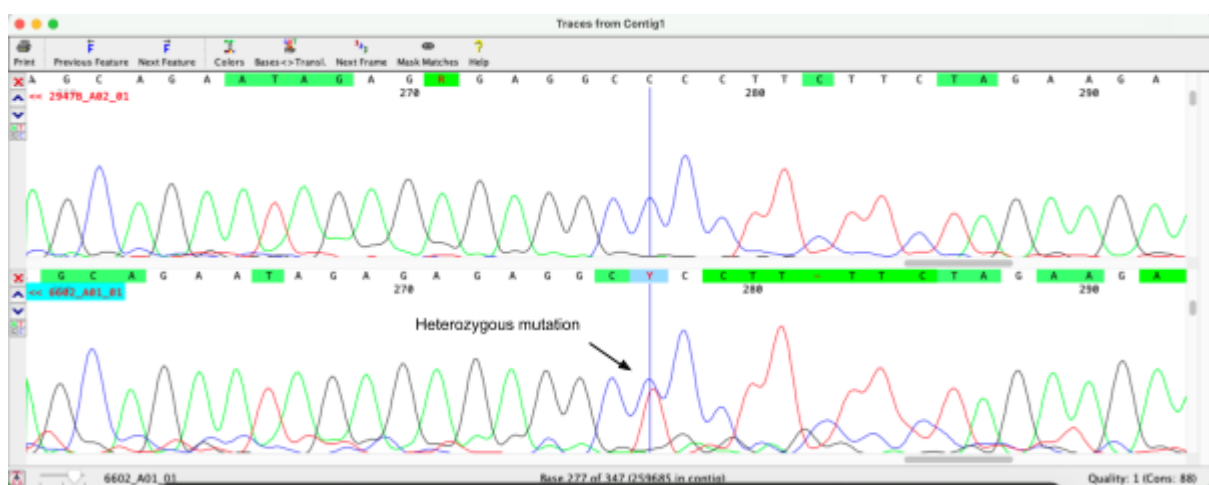


Fig 4.3.2 Representative chromatogram of the *KMT2C* gene obtained by Sanger sequencing and analyzed using CodonCode Aligner software. The chromatogram shows a heterozygous C>T missense variant in breast cancer sample GFTNBC060

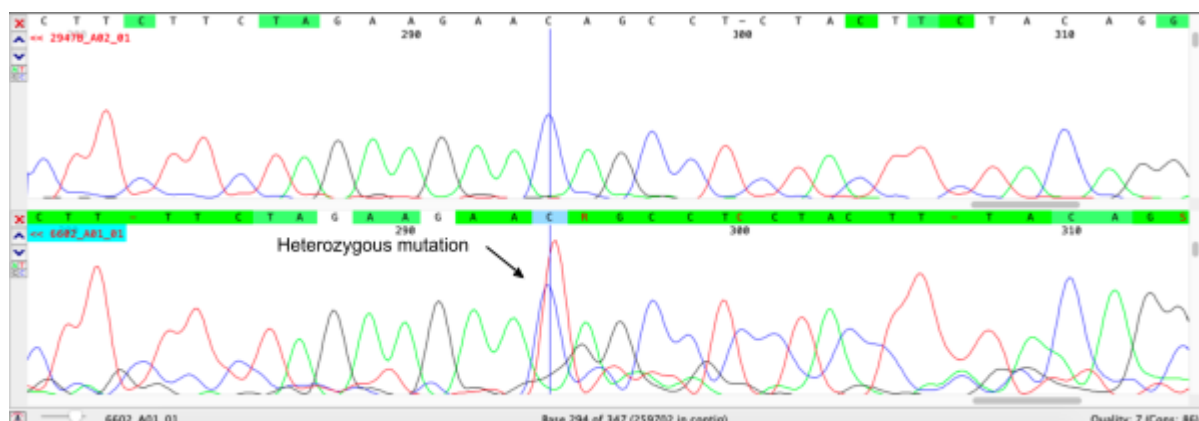


Fig 4.3.3 Sanger sequencing chromatogram of the *KMT2C* gene in breast cancer sample GFTNBC060 analyzed using CodonCode Aligner software, showing a heterozygous stop-gain variant compared with the reference (wild-type) sequence.

4.4 Identification of *KMT2C* Variants

Sequence analysis revealed that only one sample (Sample ID:GFTNBC060) contained variants in the *KMT2C* gene. Two variants were detected within this sample.

Sample ID	rsID	Variant Type	Clinical Annotation
GFTNBC060	rs760195362	Missense Variant	Likely Benign
GFTNBC060	rs2129112794	Stop-gained variant	Not reported in ClinVar

Table 4.4.1 *KMT2C* Variants Identified in Sample GFTNBC060

rs760195362		Current Build 15 Released September 3, 2015	
Organism	<i>Homo sapiens</i>	Clinical Significance	Reported in ClinVar
Position	chr7:152176319 (GRCh38.p14)	Gene : Consequence	<i>KMT2C</i> : Missense Variant
Alleles	G>A / G>C / G>T	Publications	0 citations
Variation Type	SNV Single Nucleotide Variation	Genomic View	See rs on genome
Frequency	C=0.000025 (15/595680, GnomAD_exomes) C=0.000026 (7/264690, TOPMED) C=0.000027 (4/149020, GnomAD_genomes) (+ 2 more)		

Fig 4.4.1 ClinVar database annotation of the identified *KMT2C* variant obtained from the National Center for Biotechnology Information (NCBI). The figure presents the genomic position, allele change, variation type (SNV), allele frequency, gene consequence (missense variant), and reported clinical significance of the identified variant.

rs2129112794				Current Build 157	
				Released September 3, 2024	
Organism	Homo sapiens			Clinical Significance	Not Reported in ClinVar
Position	chr7:152176302 (GRCh38.p14) ?			Gene : Consequence	KMT2C : Stop Gained
Alleles	G>A			Publications	0 citations
Variation Type	SNV Single Nucleotide Variation			Genomic View	See rs on genome
Frequency	None				

Fig 4.4.2 ClinVar database annotation of the identified KMT2C stop-gain variant obtained from the National Center for Biotechnology Information (NCBI). The figure presents the genomic position, allele change, variation type (SNV), gene consequence (stop-gain variant), and the reported clinical significance of the identified variant. .

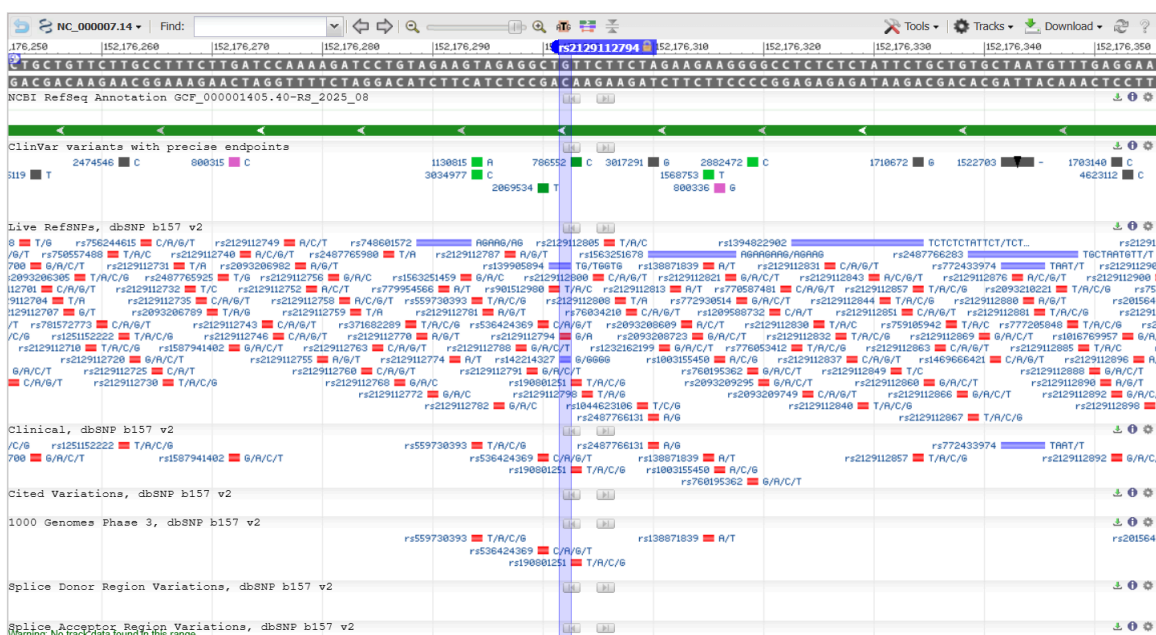


Fig 4.4.3 Genomic visualization of the rs2129112794 locus on chromosome 7 in the NCBI Genome Data Viewer, showing RefSeq annotations and overlapping variant information from ClinVar, dbSNP, and the 1000 Genomes Project.

4.5 Characterization of the Identified Variants

The first detected variant, rs760195362, was identified as a missense variant and is currently classified as likely benign. The second variant, rs2129112794, was identified as a stop-gained (nonsense) variant, which is predicted to introduce a premature stop codon in the coding sequence of the *KMT2C* gene. This variant has not been reported in ClinVar, and therefore its clinical significance remains uncertain.

4.6 Mutation Frequency

Out of the nine successfully sequenced samples, one sample harbored KMT2C variants, giving a mutation-positive frequency of 11.1%.

Parameter	Value
Total sequenced samples	8
Variant positive samples	1
Variant frequency	11.1%
Total variants detected	2

Table 4.6.1 Mutation Frequency

CHAPTER 5

DISCUSSION

The present study was undertaken to identify and characterize genetic variants in the *KMT2C* gene in breast cancer samples using PCR amplification, agarose gel electrophoresis, and Sanger sequencing. *KMT2C* is an epigenetic regulator that plays an important role in chromatin remodeling and transcriptional regulation. Alterations in this gene have been reported in several cancers, including breast cancer, making it a suitable candidate for mutation screening.

A total of 12 breast cancer samples were initially collected for the study. Following PCR amplification, one sample failed to amplify and was excluded. Among the remaining samples, three produced faint PCR bands and were therefore not suitable for Sanger sequencing. Consequently, eight samples with good-quality PCR products were sequenced. This emphasizes the importance of DNA quality and successful PCR amplification in obtaining reliable sequencing results.

Sequence analysis revealed two variants in a single sample (Sample ID: 6602). These included rs760195362, a missense variant classified as likely benign, and rs2129112794, a stop-gained variant that has not yet been reported in ClinVar. The stop-gained variant is of particular interest because it is predicted to introduce a premature stop codon, which may affect the normal function of the *KMT2C* protein. However, based on the available evidence and the findings of the present study, no conclusion can be drawn regarding its pathogenicity or its direct role in breast cancer. Further functional studies are required to determine its biological significance.

Previous studies have demonstrated that *KMT2C* is one of the frequently mutated epigenetic regulator genes in breast cancer, particularly in hormone receptor-positive tumors. Several reports suggest that loss-of-function mutations in *KMT2C* may contribute to altered gene expression, impaired DNA damage repair, and tumor progression. The identification of a stop-gained variant in the present study is therefore noteworthy, although further studies involving larger patient cohorts and functional assays are required to determine its biological and clinical relevance.

One of the major limitations of the present study was the limited sample size. Since only eight samples were successfully analyzed by Sanger sequencing, the findings cannot be generalized to the broader breast cancer population. In addition, the study was conducted within a limited internship period, which restricted the number of samples that could be collected, processed, and analyzed.

Another important limitation is that only a selected region of the *KMT2C* gene was amplified and sequenced. *KMT2C* is a large gene containing multiple exons, and variants located outside the targeted region would not have been detected using the present approach. Therefore, the absence of mutations in the remaining samples does not necessarily indicate the absence of *KMT2C* alterations throughout the entire gene.

Furthermore, Sanger sequencing was used as the mutation detection method. Although it is considered the gold standard for validating sequence variants, it is limited to the analysis of

predefined target regions and has lower throughput than Next-Generation Sequencing (NGS). Broader genomic approaches could identify additional variants and provide a more comprehensive understanding of the mutation profile of *KMT2C* in breast cancer.

Despite these limitations, the present study successfully demonstrated that PCR, agarose gel electrophoresis, and Sanger sequencing are reliable techniques for the identification of *KMT2C* gene variants. The identification of a missense variant and a stop-gained variant in one breast cancer sample contributes preliminary information on *KMT2C* genetic alterations in the studied population. Future studies involving larger sample sizes, analysis of the complete *KMT2C* gene, and functional validation of identified variants are required to clarify their biological and clinical significance. Such studies may help determine whether *KMT2C* variants can serve as useful biomarkers or therapeutic targets in breast cancer.

CHAPTER 6

CONCLUSION

The present study was carried out to identify and characterize genetic variants in the *KMT2C* gene in breast cancer samples using polymerase chain reaction (PCR), agarose gel electrophoresis, and Sanger sequencing. PCR amplification successfully generated the target DNA fragments, and agarose gel electrophoresis confirmed the quality of the amplified products prior to sequencing.

A total of 12 breast cancer samples were initially collected. After PCR quality assessment, 8 samples with satisfactory amplification were subjected to Sanger sequencing. Sequence analysis identified two variants in a single breast cancer sample (Sample ID: GFTNBC060). The detected variants included rs760195362, a missense variant classified as likely benign, and rs2129112794, a stop-gained variant that has not yet been reported in ClinVar. The stop-gained variant is predicted to introduce a premature stop codon, which may affect the normal function of the *KMT2C* protein; however, its clinical significance remains uncertain and requires further investigation.

The findings of this study demonstrate that PCR followed by Sanger sequencing is an effective approach for identifying genetic variants in the *KMT2C* gene. Although variants were detected in only one of the analyzed samples, the study provides preliminary evidence of *KMT2C* gene alterations in breast cancer. These observations contribute to the existing knowledge of *KMT2C* genetic variation and emphasize the need for further research involving larger sample sizes, comprehensive genomic analysis, and functional validation to better understand the biological and clinical significance of these variants.

Overall, this study establishes a foundation for future investigations into the role of *KMT2C* in breast cancer and highlights its potential importance in the molecular characterization of the disease.

CHAPTER 7

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